

Section : 65- A, K	Quiz 2	Time : 25 minutes
CSE- 311	DATABASE MANAGEMENT SYSTEM	

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Consider the following table: **EMPLOYEE**

EmployeeID	Name	Department	Salary	JoinDate
1	Alice	HR	50000	2020-05-01
2	Bob	IT	60000	2021-06-15
3	Charlie	IT	55000	2021-01-10
4	Diana	Marketing	45000	2020-08-25

a) Create a view named **IT_Employees** that retrieves only the employees from the "IT" department, showing EmployeeID, Name, Salary and the year of experience.

b) Write a trigger named **SalaryAudit** that logs any updates to the Salary column into a separate table SalaryChanges with the following columns: ChangeID, EmployeeID, OldSalary, NewSalary, and ChangeDate. Create the SalaryChange table and the trigger.

c) Write a stored procedure named **GiveRaise** that accepts an employee's EmployeeID and a percentage increment as input, increases their salary accordingly and updates their salary in the Employee table.

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a) Create View – IT_Employees

We need:

- Only "IT" department employees
- Columns: EmployeeID, Name, Salary, Year of Experience
- Year of Experience = difference between current year and JoinDate

```
CREATE VIEW IT_Employees AS
SELECT
    EmployeeID,
    Name,
    Salary,
    TIMESTAMPDIFF(YEAR, JoinDate, CURDATE()) AS YearsOfExperience
FROM Employee
WHERE Department = 'IT';
```

b) Trigger – SalaryAudit

Step 1: Create SalaryChanges table

```
CREATE TABLE SalaryChanges (
    ChangeID INT AUTO_INCREMENT PRIMARY KEY,
    EmployeeID INT,
    OldSalary DECIMAL(10,2),
    NewSalary DECIMAL(10,2),
    ChangeDate DATETIME
);
```

Step 2: Create the Trigger

```
DELIMITER $$

CREATE TRIGGER SalaryAudit
BEFORE UPDATE ON Employee
FOR EACH ROW
BEGIN
    IF NEW.Salary  OLD.Salary THEN
        INSERT INTO SalaryChanges (EmployeeID, OldSalary, NewSalary,
ChangeDate)
        VALUES (OLD.EmployeeID, OLD.Salary, NEW.Salary, NOW());
    END IF;
END$$

DELIMITER ;
```

c) Stored Procedure – GiveRaise

```
DELIMITER $$

CREATE PROCEDURE GiveRaise(
    IN empID INT,
    IN percentage DECIMAL(5,2)
)
BEGIN
    UPDATE Employee
    SET Salary = Salary + (Salary * percentage / 100)
    WHERE EmployeeID = empID;
END$$

DELIMITER ;
```

Calling Example:

```
CALL GiveRaise(2, 10); -- Increase Bob's salary by 10%
```

☒ This fully answers:

- **a)** View creation
 - **b)** Trigger with separate change log table
 - **c)** Stored procedure to give raises
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